

LESSON 13.2

JUSTIFYING MEASURES OF CENTER AND SPREAD



LESSON INTRODUCTION

MA.7.DP.1.1 Determine an appropriate measure of center or measure of variation to summarize numerical data, represented numerically or graphically, taking into consideration the context and any outliers.

LEARNING TARGETS

- I can justify whether or not a measure of center or variation appropriately summarizes data.

BENCHMARK COHERENCE

BUILDING ON

MA.6.DP.1.2 Given a numerical data set within a real-world context, find and interpret mean, median, mode, and range.

MA.6.DP.1.6 Given a real-world scenario, determine and describe how changes in data values impact measures of center and variation.

WORKING TOWARD

MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical data, and on whether it is univariate or bivariate.

MA.912.DP.1.2 Interpret data distributions represented in various ways. State whether the data is numerical or categorical, whether it is univariate or bivariate, and interpret the different components and quantities in the display.

ESSENTIAL QUESTIONS

- How do you determine which measures of center best describe a data set?
- How do you determine which measures of variation best describe a data set?

LESSON NARRATIVE

In this lesson, students build on their understanding of the effect of outliers from Lesson 1. Students begin by using histograms and stem-and-leaf plots to analyze data and consider which measures of center and variation make the most sense to describe sets of data. Students then draw conclusions about how outliers or the shape of a distribution impact the mean, median, range, and IQR and apply this knowledge throughout the rest of the lesson to determine which measures are best to describe data sets given the context of the data presented.

COMPONENT SUMMARY

13.2.1 WARM-UP (~5 MIN.)

Notice and Wonder about income in different countries

13.2.3 BRING IT TOGETHER (5 MIN.)

Describe the effects of outliers on measures of center and variation to determine which measures to use based on a data set

13.2.5 REFLECTION (~5 MIN.)

Self-assessment of confidence in the day's learning targets

13.2.2 COLLABORATION (15-20 MIN.)

Describe and interpret measures of center and spread from data displayed in histograms and stem-and-leaf plots

13.2.4 GUIDED PRACTICE (10 MIN.)

Practice determining which measure of center and variation best summarizes a data set

RESOURCES

GLOSSARY

Key Terms:

- Median
- Interquartile range

Additional Terms:

- Mean
- Range

MATERIALS

- Calculators

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle to estimate the mean and median and determine which will be larger based on the shape of a distribution.
- Students may not understand which measure of center or variation best describes a given data set.
- Students may struggle to connect context to the measures of center and variation.

THINGS TO CONSIDER

- Consider encouraging students to refer back to the graphs and data displays to help visualize the data. Considering the shape of the distribution is helpful when determining the most appropriate measures of center and spread.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

13.2.2 Collaboration provides an opportunity to engage students in a Think-Pair-Share routine as they analyze student responses to data. Have each student think individually about the problem, then discuss it with their partner. As students talk, circulate to find 1-2 responses to share with the class that will advance student thinking and learning.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD**MA.K12.MTR.7.1 Real-World Contexts**

This lesson provides a multitude of opportunities for students to engage in discourse that connects mathematics to real-world contexts. Students consider how the context of the data set impacts which measures of center and variation are most appropriate to describe the data.

DIFFERENTIATE INSTRUCTION

Throughout this lesson, visual entry points are provided by displaying data using various displays such as line plots and histograms. Auditory entry points can be provided through partner and class discussions.

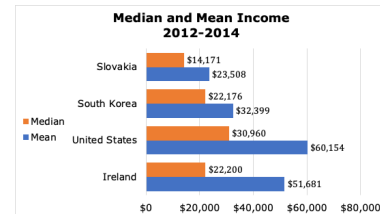
13.2.1 WARM-UP: INCOME**ENRICHMENT**

This Warm-Up positions students to consider mean and median in a real-world context. This sets students up to consider how context can impact the ways in which data can/should be interpreted and which measure of center/variation makes the most sense to consider. To connect to the rest of the lesson, ask questions like:

- Which measure of center would you want to use to best describe the average income in the U.S.? Why?
- What does this tell us about the data? Why is that important? Does this apply to all data sets? Why or why not?

**13.2.1 Warm-Up: Income**

The diagram below shows bar charts for the median and mean income between 2012 and 2014 in selected countries. The income is given in U.S dollars (\$).

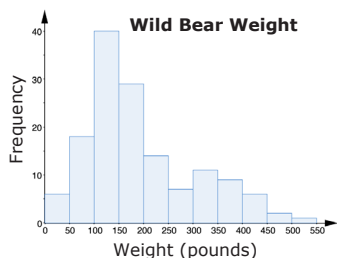


1. What do you notice about the values for median and mean income?
Answers will vary. The mean income values are higher than the median.
2. Why do you think the mean income for the U.S. was so much higher than the median income?
Answers will vary. There could be outliers in the data, and outliers affect the mean more than the median.



13.2.2 Collaboration: Center and Variation in Context

1. In a study of wild bears, researchers measured the weights, in pounds (lb), of 143 wild bears that ranged in age from newborn to 15 years old. The data was used to make the histogram shown.



- a. Complete the statements.

Based on the data, it appears wild bears typically

weigh between
☐ 50 and 150
☒ 100 and 200
☐ 150 and 250
 pounds. A good

estimate for the typical weight for wild bears is 175

pounds.

- b. Compare your estimate with your partner. Discuss whether it is possible for both of your estimates to be reasonable, even if they are not the same.

- c. Discuss with your partner whether your estimate from Part A is closer to the median or the mean. Summarize your discussion.

Answers will vary. The estimate would be closest to the median. The few bears in the data set that weigh more will cause the mean to be higher than the median.

- d. Angelica and Raymond were asked to describe the variation in weight of the wild bears.

Angelica said, "The bear weights range from less than 50 to over 500 lbs, so the range best describes the spread of the weights of the wild bears."

Raymond said, "It looks like most bears in the sample weigh between 100 and 300 lbs, with few weighing less and few weighing more. The small weights could be because there were babies in the sample. So, the interquartile range is the best measure of spread to describe this data."

Explain whether you agree with Angelica or Raymond. Justify your reasoning.

Answers will vary. Raymond's explanation best describes the spread. He justifies his choice with the understanding that the small weights are because the bears were babies and the frequency for the weights between 100 and 300 is much higher than at heavier weights.

13.2.2 COLLABORATION: CENTER AND VARIATION IN CONTEXT

ENRICHMENT

Students may struggle to understand what a "typical weight" is in this context. Consider asking guiding questions like: • Between what numbers do most of the weights fall in the data set?

- What does typical mean in this context? Is it referring to the center or variation of the data set?
- If the most weights are between 100-150 and 150-200, how can you describe the typical weights of the wild bears?

DIFFERENTIATE INSTRUCTION

Auditory entry points are built in throughout this component through partner discussion. Consider providing additional visual entry points through the use of diagrams. Encourage students to use colored pencils to mark the median and mean on the histogram to help visualize their estimations and response.

TEACHER MOVES

Think-Pair-Share

Consider implementing a Think-Pair-Share routine for question 1d. Have each student think individually about the problem, then discuss it with their partner. As students talk, circulate to find 1-2 responses to share with the class that will advance student thinking and learning. Listen for students who can justify why the IQR makes the most sense based on the shape of the distribution.

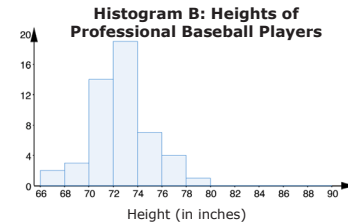
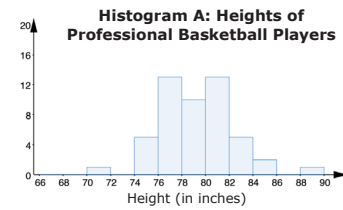
13.2.2 COLLABORATION: CENTER AND VARIATION IN CONTEXT (CONTINUED)

ENRICHMENT

Students may easily guess the answers to question 2. However, connect it to the next activity by asking questions like:

- What values appear to be outliers?
- What would the mean and median be of each histogram?
- Is the mode easiest to find when given a histogram?
- On average are basketball players or baseball players taller?

2. The two histograms, labeled A and B, show height distributions of 50 male professional basketball players and 50 male professional baseball players.



Work with your partner to complete each statement.

- a. In ☐ Histogram A, ☒ Histogram B, the average height is around 72 inches. The values of the mean and median are similar.
- b. To explain typical height, the variation in ☒ Histogram A ☐ Histogram B is best described by the interquartile range because the range may be misleading due to outliers. The values of the mean and median are similar.

ENRICHMENT

If students are struggling to answer question 3b, provide additional scaffolding through intentional questions like:

- What do you notice about the spread of the data? Why is this important when considering measure of center?
- Based on what you know so far about measures of center, how can outliers impact the mean, or average?
- Is there one measure of center that would be more accurate to describe the data? Why?

3. An earthworm farmer set up several containers of a certain species of earthworm so he could learn about their lengths. Lengths of earthworms provide information about their ages. The farmer measured the lengths of the earthworms in one container.

The farmer displayed the earthworms' lengths, in millimeters (mm), in a stem-and-leaf plot.

Stem	Leaf
0	6
1	1, 8, 9
2	0, 3, 3, 5, 5, 6, 7, 7, 8, 9
3	2, 3
4	1, 2, 8
5	2, 4, 9
6	0
7	7
8	
9	3

Key: 2|8 = 28

Use the display to answer the following questions.

- a. Half of the earthworms sampled were less than 28 mm long.
- b. The farmer reported that the average length is 35.92 mm. Why might this measure of center be misleading about the typical length of the earthworms in this sample?
Answers will vary. This could be misleading because of the vast range of data.
- c. Compare your answers to Parts A and B with your partner and, if necessary, resolve any differences.

- d. The farmer reported that the lengths varied between 6 mm and 93 mm. The farmer's assistant said that most earthworms had lengths between 23 and 50 mm.

Work with your partner to complete the statements.

The farmer is using the ☐ IQR
☒ range to summarize

the variation in worm length, while the farmer's

assistant is using the ☒ IQR.
☐ range.

If a researcher is interested in how long

earthworms are, the ☒ IQR
☐ range is an appropriate

measure of variation for the farmer to report

because

Answers will vary. IQR is not affected by outliers.

13.2.2 COLLABORATION: CENTER AND VARIATION IN CONTEXT (CONTINUED)



13.2.3 Bring It Together: Outlier Effects

1. Complete each statement to describe the effect outliers have on each measure of center or variation.

- The presence of an outlier does not have a significant effect on the ☐ mean.
☒ median.
- The presence of an outlier will change, often significantly, the value of the ☒ mean.
☐ median.
- The presence of an outlier does not have a significant effect on the ☒ IQR.
☐ range.
- The presence of an outlier will change, often significantly, the value of the ☐ IQR.
☒ range.



When outliers are present or a distribution is skewed, the **median** is the preferred measure of center and the **IQR** is the preferred measure of variation to describe the distribution. However, the context must be considered because outliers may be important in understanding the data.

13.2.3 BRING IT TOGETHER: OUTLIER EFFECTS

TEACHER MOVES

Students have informally worked through these concepts in previous components of this lesson and in Lesson 1 but are now drawing formal conclusions about how outliers impact measures of center and variation. Review the blue box with the class to drive home the point that context matters when considering whether or not to include outliers in a data set. Consider providing an example:

- Outliers can be in the analysis of large data sets because they are less likely to significantly impact the data.
- In the real world, it would be important to include outliers in data sets for medical trials because they may impact the results of the trial.

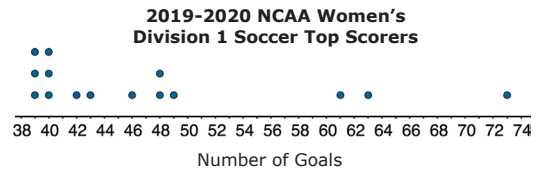
Ensure students understand that even when outliers are not present, if the shape of the distribution is skewed, it is still best to use the median and IQR to describe the distribution for a more accurate description of the data set.

13.2.4 GUIDED PRACTICE: COMPARING DISTRIBUTIONS



13.2.4 Guided Practice: Comparing Distributions

- Below is a line plot of the number of goals scored by the top 15 women goal scorers during the 2019-2020 NCAA Women's Division 1 soccer season.



Rank	Player, School	Number of Goals
1	Evelyne Viens, South Florida	73
2	Catarina Macario, Stanford	63
3	Raimée Sherle, Boise St.	61
4	Ally Watt, Texas A&M	49
5	Tziarra King, NC State	48
5	Deyna Castellanos, Florida St.	48
7	Alexa Genas, Campbell	46
8	Morgan Weaver, Washington St.	43
9	Peyton DePriest, Middle Tenn.	42
10	Elise Flake, BYU	40
10	Samantha Dewey, Xavier	40
10	Hayley Younginer, Wake Forest	40
13	Dakota Mills, Saint Joseph's	39
13	Katie McClure, Kansas	39
13	Taylor Kornieck, Colorado	39

ENRICHMENT

If you notice students struggling to explain which measure of central tendency will be larger, encourage them to use the line plot to help visualize. Ask prompting questions like:

- What do you notice about the data? Do there appear to be any outliers? Which measure of center do outliers impact more?
- How will the outliers impact the mean? The median? How can you use this to determine which will be larger?

TEACHER MOVES

Allow students to use calculators to find the mean number of goals.

TEACHER MOVES

Students may not understand which measure of variation should be used to summarize the data if they don't have a strong grasp of why context matters when interpreting data. Guide students toward finding the range and the IQR if they are struggling to answer. Then, have them refer back to the line plot to determine which measure is more representative of the data.

- Without doing any calculations, explain which measure of central tendency you expect to be larger, the mean or the median.
Answers will vary. The mean will be larger because 73 will skew the data.
- Find the mean number of goals.
47.3
- Find the median number of goals.
43
- Check your work with your partner, and if necessary, resolve any differences.
- Demetricka stated that the typical number of goals in the season for a scorer in the top 15 was about 48.

Explain whether you agree with Demetricka using your work from Parts A-C.

Answers will vary. According to the mean of the data, Demetricka's estimate would be correct. However, according to the median, Demetricka's estimate would be high.

- Demetricka described the variation by saying that most of the players in the top 15 had around the same number of goals scored, with only a few scoring much higher.
 - Given her description, discuss with your partner whether Demetricka should use the range or the interquartile range to numerically summarize the spread of the number of goals scored.
Answers will vary. The range of the data is 34 and the IQR is 9.
 - Explain which measure of variation is most appropriate to summarize the data.
The IQR would better summarize the spread of the data because it eliminates the effects of the few higher numbers.

3. At times, a statistical purpose may justify using a particular measure to describe the data.

Complete the statement.

If the statistical purpose of collecting the data is to

determine how high the very top scorers score, the

☐ IQR
☒ range would be the most appropriate measure

to use because this information is not captured in the

☒ IQR.
☐ range.

13.2.4 GUIDED PRACTICE: COMPARING DISTRIBUTIONS (CONTINUED)



13.2.5 Wrap-Up: Reflection

Draw an **X** on the line to indicate where your current learning is in relation to the following learning targets.

1. I can justify whether or not a measure of center appropriately summarizes data.
Answers will vary.

I need help.
I can't get started. I'm getting there, but
I need more practice. I understand this
and I feel confident.



2. I can justify whether or not a measure of variation appropriately summarizes data.
Answers will vary.

I need help.
I can't get started. I'm getting there, but
I need more practice. I understand this
and I feel confident.



13.2.5 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.